

Problem Solving

Mathematics 1000

Summer, 2018

1 BASIC INFORMATION

Lecture: MTWRF 9:10 am - 11:20 am, 5/21 - 6/15, EN 3104

Lecturer: Bryce Christopherson **Email:** bchris19@uwyo.edu

Office Location: Ross Hall 303 **Office Hours:** MTWR 12:15 pm - 1:15 pm

2 COURSE DESCRIPTION:

This course satisfies the Quantitative Reasoning 1 (QA) requirement of the 2003 USP, and the Quantitative Reasoning (Q) requirement of the 2015 USP.

3 PREREQUISITE INFORMATION:

The (official) prerequisite for this class can be satisfied with any of the following:

- ACT Math score of 21 or better within the last three years.
- SAT Math score of 600 or better within the last three years.
- Math Placement Exam score (Compass, AccuPlacer, or ALEKS PPL) of Level 2 within the last year.
- A grade of C or better in Math 0921 or higher.

Subsequent QB courses are STAT 2000/2050/2070. Students can also progress to Math 1100.

This course is not a prerequisite for Math 1400 College Algebra.

4 COURSE MATERIALS

4.1 SOFTWARE (REQUIRED):

MyMathLab (includes access to the e-book). If you have an old access code that you have used in the past with this edition of the book and it is not older than one year, you can still use it. You will enroll in MyMathLab using an access code (access code can be purchased from the university bookstore) through the shell in WyoCourses. Details on how to enroll for MyMathLab will be posted on WyoCourses. **For best results, purchase your access code NOW and start working on your homework from the first day of class.**

4.2 INTERNET ACCESS (REQUIRED):

You will need it to use the MyMathLab software to finish your online homework.

4.3 PHYSICAL TEXTBOOK (OPTIONAL):

We will use the textbook *Excursions in Modern Mathematics* by Peter Tannenbaum, 8th Edition. There is a student solutions manual available for this text. The only required portion is the online access package, available at the UW Bookstore or directly through Pearson. If you buy your access code from the university bookstore, you can get the physical copy of the book at a lower cost if you purchase them as a bundle.

4.4 CALCULATOR (REQUIRED):

A scientific non-graphing calculator is strongly recommended for this course. During exams and quizzes, you will not be allowed to use a graphing calculator or a cell phone and you will not be allowed to share a calculator with someone else.

4.5 NOTE CARDS (OPTIONAL):

For each exam you will be allowed to use both sides of one 3x5 index card for hand-written notes.

5 GRADING

The grade you receive at the end of this course will be based on your performance in several categories, which determine your final grade percent according to the table on the left. The approximate grading scale used to evaluate this total percentage is given in the table on the right.

Assignment Group	Grade Percentage	Letter Grade	Overall Percentage
Exams	60%		
Exam 1	15%	A	[90,∞)
Exam 2	15%		
Exam 3	15%	B	[80,90)
Exam 4	15%		
MyMathLab	20%	C	[70,80)
Quizzes	20%	D	[60,70)
Total	100%	F	[0,60)

6 EXAMS (60%)

Exams 1-4 will be given during the regularly scheduled class time (9:10 am -11:20 am) on the dates shown below. On each exam, keep the following in mind for best results:

- (i) On any problem that requires a computation, show your work and reasoning in an organized, logical, reasonable fashion.
- (ii) Before and after you write an answer, re-read the problem to make sure you have answered the question it is asking.
- (iii) Don't use the exam to explain why you aren't doing well on the exam – please talk to me instead (preferably ahead of time).

6.1 EXAM 1

Friday, 5/25 Covering sections 5.2, 5.3, 6.2, 6.4

6.2 EXAM 2

Friday, 6/1 Covering sections 7.1, 7.2, 7.3, 17.1, 17.2, 17.4

6.3 EXAM 3

Friday, 6/8 Covering sections 10.1, 10.2, 10.3

6.4 EXAM 4

Friday, 6/15 Covering sections 10.4, 13.1, 13.2

6.5 MAKE-UP EXAMS

If you will not be able to take an exam on the scheduled date and time, you **must** make other arrangements with the instructor **well in advance**. In cases where proper notification has not been given and the reason for missing the exam is not valid, a make-up the exam **will not be given**.

7 MYMATHLAB HOMEWORK (20%)

The instructions for registering for MyMathLab (MML), can be found on the WyoCourse site. Online assignments should usually take around 30 minutes to complete, on average. These problems are chosen as representative of the basic concepts presented during lecture. It is hoped that you will complete the MML problems before the due date so that these problems may prompt discussion in the next lecture.

For full credit, MML assignments **must be completed by 11:59pm on their due date**. MML problems can be completed after the due date for half credit. You have an unlimited number of attempts on each problem, so you should strive for nothing less than 100% on each online homework assignment. None of the MML assignments will be dropped. Check the attached course schedule or the MML website for the assignment due dates.

8 QUIZZES (20%)

A short quiz will be given at the end of a number of classes (see schedule). The quizzes will cover the sections indicated in the schedule and will be similar to the MML homework. You are allowed to use a non-graphing calculator on each quiz. None of the quiz scores will be dropped. If you miss a quiz and have not made prior arrangements with the instructor to account for your absence, a make-up quiz **will not be given**.

9 EXTRA CREDIT

Opportunities for earning extra credit may occur throughout the semester at the discretion of the instructor. Extra credit problems may be given in class (Note: such problems will not be emailed to anyone not in class at the time) or occasionally may be offered on quizzes and exams. Points earned from extra credit may help borderline grades, but do not expect extra credit to significantly change your final score.

10 EMAIL CORRESPONDENCE

Please always put Math 1000 in the subject line of any email you send to me. This will help me spot your email more easily, which translates to a quicker response. An email should have a salutation, body, and signature.

11 ATTENDANCE

Attendance to regularly scheduled lecture sessions is not mandatory, but I believe it is very important. If you frequently do not attend lecture without an exceptionally good reason and begin to struggle in the course, I will have very little sympathy for you.

12 STUDY TIPS

Make sure you come to each class prepared. Engage with the material during lecture – strive for genuine and deep understanding, not just a grade. Spend at least two hours outside of class on a subject for each hour in class.

Form study groups so you can discuss the contents of the course with other members of our class. Visit me! During the office hours I am required to hold, my first priority is to help students with their questions. I am also available via email.

You can seek out for FREE tutoring in the Math Assistance Center (MAC). The MAC, located in Ross Hall 29, offers free mathematics assistance for 1000 and 2000 level math classes. No appointment is required, just drop in. You can even go there to do your homework and you do not have to ask any questions or to seek any help. It is also a good place to meet your group mates if you are working on an assignment together. The MAC has 9 desktop computers that you can use to do your online homework in addition to a smart board/projector. You can find current textbooks, mock exams, cheat sheets, and other helpful study resources. Of course, in addition to the friendly helpful staff!

The MAC website is <http://www.uwyo.edu/math/mac/>

13 ACADEMIC HONESTY

The University of Wyoming is built upon a strong foundation of integrity, respect and trust. All members of the university community have a responsibility to be honest and to expect honesty from others. Cheating, plagiarism, and other forms of academic dishonesty are unacceptable. Students should report suspected academic dishonesty to the instructor. This course follows the recommendations of the College of Arts and Sciences in addition to the University Regulation 6-802 and the Student Code of Conduct. The UW regulations and the Student Code of Conduct (UW Regulation 8-30) can be found at the Office of General Counsel page on the UW website.

14 DISABILITY ACCOMMODATION

If you have a physical, learning, sensory or psychological disability and desire accommodations, please let your instructor know as soon as possible. You will need to register with University Disability Support Services (UDSS), room 109 Knight Hall. You may also contact UDSS at (307) 766-6189, udss@uwyo.edu, or visit the UDSS page of the UW website for more information.

If you feel that you have what people refer to as math anxiety, I urge you to call or visit the Counseling Center in room 341, Knight Hall and get some help. Their phone number is 766-2187. Many people with math anxiety tend to score lower on math exams, which is something that may reduce your chance to pass this course.

15 CHANGES TO THE SYLLABUS

Substantive changes made to the syllabus by the instructor during the semester shall be communicated in writing to the students.

16 GRADING RUBRIC

The rubric on the following page will be used as a guideline when evaluating your written work on quizzes and exams throughout this course. You should strive to produce written solutions that are clear and well organized, make proper use of mathematical notation and terminology, demonstrate an understanding of the mathematical concepts involved, and result in a correct solution. If you practice these skills while working on your MML problems, you will be more likely to meet the standard expected of your written work on quizzes and exams.

Please familiarize yourself with this rubric and reference it if ever you feel your work has been unfairly graded.

Characteristics of the Written Work	Qualitative Description	Typical Scoring %
(i) The work is presented in a neat, clear, organized fashion that is complete and easy to follow. (ii) Graphs are correct, complete, carefully drawn and appropriately labeled. (iii) Correct terminology and notation are always used. (iv) The solution shows a complete understanding of the mathematical concepts involved, although it may contain a minor arithmetic error or sign mistake.	Exceptional	90-100
(i) The work may not be completely clear or well organized, or it may have some steps missing, but it is still possible to follow with little effort. (ii) Graphs are correct and complete, but somewhat carelessly drawn or missing some labels. (iii) Correct terminology and notation are usually used. (iv) The solution shows almost complete understanding of the mathematical concepts and may contain one or two arithmetic or sign errors.	Good	75-90
(i) The work may be unclear, poorly organized, or have several steps missing, but it is possible to deduce the intended method of solution with some effort. (ii) Graphs are partially correct, somewhat carelessly drawn, and may be missing some labels. (iii) Correct terminology and notation are sometimes used. (iv) The solution shows a limited understanding of the mathematical concepts involved, or an almost complete understanding of the mathematical concepts involved, but contains several arithmetic or sign errors.	Fair	60-75
(i) The work is poorly organized or is very incomplete, making it difficult to follow. (ii) Graphs are incorrect, carelessly drawn, or incomplete, and may not be labeled. (iii) Correct terminology and notation are seldom used. (iv) The solution shows a limited understanding of the mathematical concepts involved, and may also contain several arithmetic or sign errors.	Poor	40-60
(i) The work is so disorganized or incomplete that it is very difficult or impossible to follow. (ii) Graphs are not drawn or are completely wrong and not labeled. (iii) There is little or no correct usage of terminology and notation. (iv) The solution shows little or no understanding of the mathematical concepts involved.	Very Poor	0-40

17 COURSE SCHEDULE

Sections Covered:

5.1,5.2,5.3,6.2,6.4,7.1,7.2,7.3,10.1,10.2,10.3,10.4,13.1,13.2,17.1,17.2,17.4

Su.	Mo.	Tu.	We.	Th	Fr.	Sa.
5/20	5/21	5/22	5/23	5/24	5/25	5/26
	5.1 5.2 5.3 MML 5.2	6.2 6.4 MML 5.3	7.1 7.2 7.3 MML 6.2	Review Quiz 1 (5.2,5.3,6.2,6.4) MML 6.4	EXAM 1 (5.2,5.3,6.2,6.4)	
5/27	5/28	5/29	5/30	5/31	6/1	6/2
MML 7.1	Memorial Day (No class) MML 7.2	17.1 Quiz 2 (7.1,7.2,7.3) MML 7.3	17.2 MML 17.2	17.4 Review MML 17.4	EXAM 2 (7.1,7.2,7.3 17.1,17.2,17.4)	
6/3	6/4	6/5	6/6	6/7	6/8	6/9
	10.1 MML 10.1	10.2 MML 10.2	10.3 MML 10.3	Review Quiz 3 10.1,10.2,10.3	EXAM 3 (10.1,10.2,10.3)	
6/10	6/11	6/12	6/13	6/14	6/15	6/16
	10.4 MML 10.4	13.1 MML 13.1	13.2 MML 13.2	Review Quiz 4 (10.4,13.1,13.2)	EXAM 4 (10.4,13.1,13.2)	